

Q. Prove that $\cos \tan^{-1} x \sin \cot^{-1} x = \sqrt{\frac{x^2+1}{x^2+2}}$

Q. Prove that $\tan \frac{\theta}{2} = \sqrt{\frac{1-e}{1+e}} \tan \phi$, prove that

$$\cos \phi = \frac{\cos \theta - e}{1 - e \cos \theta}$$

2013

Q. Prove that:

$$(a) \cos 24^\circ + \cos 55^\circ + \cos 125^\circ + \cos 204^\circ + \cos 306^\circ = \frac{1}{2}$$

$$(b) \frac{\sin(A-B)}{\sin A \cdot \sin B} + \frac{\sin(B-C)}{\sin B \cdot \sin C} + \frac{\sin(C-A)}{\sin C \cdot \sin A} = 0$$

Q. If $2 \tan A = 3 \tan B$, show that

$$\tan(A-B) = \frac{\sin 2B}{5 - \cos 2B}$$

$$Q. \sin^{-1} \frac{3}{5} + \sin^{-1} \frac{8}{17} = \sin^{-1} \frac{77}{85}$$

2014

Q. Prove that:

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1} [x \sqrt{1-y^2} + y \sqrt{1-x^2}]$$

Q. Prove that $2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} = \pi/4$

Q. If $A+B=45^\circ$, show that $(1+\tan A)(1+\tan B) = 2$

Prove that:

$$\sin 12^\circ \cdot \sin 48^\circ \cdot \sin 54^\circ = \frac{1}{8}$$

2015

Q. Prove that:

$$\cos 20^\circ \cdot \cos 40^\circ \cdot \cos 60^\circ \cdot \cos 80^\circ = \frac{1}{16}$$

Q. If $\tan \theta = \frac{a}{b}$ then prove that $a \sin 2\theta + b \cos 2\theta = b$.

Q. Prove that: $2 \tan^{-1} x = \sin^{-1} \frac{2x}{1-x^2} = \cos^{-1} \frac{1-x^2}{1+x^2} = \tan^{-1} \frac{2x}{1-x^2}$

Q. If any $\triangle ABC$, show that:

$$\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$$

2016

Q. Prove that $\cos x \cos 2x \cos 4x \cos 8x = \frac{\sin 16x}{16 \sin x}$

Q. Prove that $2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} = \frac{\pi}{4}$

2017

Q. Prove that $\sin 10^\circ \cdot \sin 30^\circ \cdot \sin 50^\circ \cdot \sin 70^\circ = \frac{1}{16}$

Q. Show that $\sin 18^\circ = \left(\frac{\sqrt{5}-1}{4} \right)$

Q. Prove that $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{8} = \frac{\pi}{2}$

2018.

Q. The ^{mc} principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ is equal to

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $-\frac{\pi}{6}$ (d) none of the above

Q. Prove that $\tan^{-1}\frac{2}{5} + \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{12} = \frac{\pi}{4}$

Q. If $A+B = 45^\circ$ then prove that $(\cot A - 1)(\cot B - 1) = 2$.

Q. Prove that in a triangle ABC, $\tan$

$$\frac{A-B}{2} = \frac{b-c}{a+b} \cot \frac{C}{2}$$

$$\frac{A-B}{2} = \frac{a-b}{a+b} \cot \frac{C}{2}$$

Q. In a triangle ABC, Prove that $a^2 \cos C - c \cos B = b^2 - c^2$

2019.

Q. Prove that $2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7} = \frac{\pi}{4}$.

Q. If $\tan A = \frac{5}{6}$ and $\tan B = \frac{1}{11}$, Prove that $A+B = 45^\circ$

Q. Prove that $75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$

NEP-2024

Q. Find the values of trigonometric functions of 120° .

Q. Prove that: $\cos \alpha + \cos \beta + \cos \gamma + \cos (\alpha + \beta + \gamma)$
 $= 4 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\beta + \gamma}{2} \right) \cos \left(\frac{\gamma + \alpha}{2} \right).$

NEP-2023

Q. In any triangle ABC, prove that $\tan \frac{B-C}{2}$
 $= \frac{b-c}{b+c} \cot \frac{A}{2}$

Q. Prove that $2 \tan^{-1} \sqrt{x} = \sec^{-1} \frac{1+x}{1-x}$

Q. If $f(x) = \tan^{-1} x$ prove that $f(x) - f(y) = f\left(\frac{x-y}{1+xy}\right)$

Q. Find the differential coefficient of $\sin^{-1} x$ from the first principle.

2022

Q. Find the value of $\cos 220^\circ + \cos 100^\circ + \cos 20^\circ$.

Q. Prove that $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$

Q. Find the value of x , if $\tan^{-1}(2x) + \tan^{-1}(3x)$

$= \frac{\pi}{4}$.
Q. Find the value of $\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 60^\circ \cdot \sin 80^\circ$.

6. (a) Prove that $\tan^{-1} \frac{2}{5} + \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{12} = \frac{\pi}{4}$.

(b) If $A + B = 45^\circ$ then prove that $(\cot A - 1)(\cot B - 1) = 2$.

7. (a) Prove that in a triangle ABC , $\tan \frac{A-B}{2} = \frac{a-b}{a+b} \cot \frac{C}{2}$.

(b) In a triangle ABC , Prove that $a(b \cos C - c \cos B) = b^2 - c^2$.

6. (a) Prove that $2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} = \frac{\pi}{4}$.

(b) If $\tan A = \frac{5}{6}$ and $\tan B = \frac{1}{11}$, prove that $A + B = 45^\circ$

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7. (a) In any $\triangle ABC$, if a^2, b^2, c^2 are in A.P., prove that $\cot A, \cot B, \cot C$ are in A.P.

(b) Prove that $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$

(a) Find the value of $\cos 220^\circ + \cos 100^\circ + \cos 20^\circ$

(b) Prove that $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$

(c) In a triangle ABC the sum of two sides is $\sqrt{3}$ times their difference and angle included between them is 60° . Find the difference of remaining angle.

6. (a) Find the value of x , if $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$.

4. (a) Find the values of all trigonometric functions of 120° .

(b) Prove that: $\cos \alpha + \cos \beta + \cos \gamma + \cos(\alpha + \beta + \gamma) = 4 \cos\left(\frac{\alpha+\beta}{2}\right) \cos\left(\frac{\beta+\gamma}{2}\right) \cos\left(\frac{\gamma+\delta}{2}\right)$. 7+7

(a) In any triangle ABC prove that $\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$.

(b) Prove that $2 \tan^{-1} \sqrt{x} = \sec^{-1} \frac{1+x}{1-x}$.

4. (a) If $\cos \theta = \frac{4}{5}$ and $\cos \varphi = \frac{12}{13}$ where θ and φ both lie in the fourth quadrant, find the value of $\tan(\theta + \varphi)$.

(b) Prove that $\frac{\sin 3x - \sin x}{\cos x - \cos 3x} = \cot 2x$

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7. (a) In any $\triangle ABC$, if a^2, b^2, c^2 are in A.P. , prove that $\cot A, \cot B, \cot C$ are in A.P.

(b) Prove that $\sin 75^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}$.